

Atharva Dalvi

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EDUCATION

Bachelor of Science in Computer Science, Texas Tech University

May 2025

Data structures, object-oriented programming, design & analysis of algorithms, software architecture, artificial intelligence (AI), concepts of database systems, computer architecture, theory of automata

SKILLS

- Programming languages: Bash, C, C++, HTML/CSS, Java, JavaScript, Python, SQL, TypeScript
- Tools: Docker, Eclipse, Git, IDLE, Linux, Microsoft Office suite, MongoDB, Visual Studio, Visual Studio Code
- Software Engineering: Angular, Dot Net, Flask, LLM, Next JS, Node JS, RESTful API, Django
- Interpersonal Skills: Customer service, Problem solving, Collaborative Teamwork

WORK EXPERIENCE

Research Engineer, Texas Tech University

August 2025 – Present

- Developing an LLM based RAG system for generating adaptive, context-aware feedback for student assignments
- Built a CLI based prototype using qwen2.5 model to generate pedagogical feedback with automatic output logging
- Assessed prompt engineering techniques and reasoning-driven prompts to enhance feedback precision
- Implemented rule-based post-generation validators to enforce output constraints and ensure structural correctness of LLM-generated feedback
- Implemented a GitHub Actions based CI pipeline to automatically validate CLI functionality, prompt construction logic, and post-generation validators through unit tests and mocked LLM API calls

Full Stack Development Intern, Respirer Living Sciences

May 2024 – July 2024

- Designed & developed a web application to monitor air quality, delivering real-time environmental metrics which boosted HVAC operational efficiency by triggering alerts to end users
- Engineered a Python/Flask API to integrate data from 20+ air quality monitors incorporating existing REST APIs
- Built a React/Node.js application utilizing D3.js and Recharts for interactive data visualization and user engagement
- Containerized the application with Docker which streamlined the deployment processes
- Deployed the solution with Nginx and Gunicorn to improve request handling performance and production reliability
- Collaborated with a team of two architectural designers to plan the dashboard layout and UX flow
- Used Figma to prototype the user interface, ensuring alignment with both functional and visual design goals

Software Development Intern, TOSBS

October 2020 – January 2021

- Developed Python scripts to analyze stocks; applied linear regression models to recommend investment strategies
- Built predictive trading models using Python libraries such as NumPy, Pandas, and Scikit-learn for stock forecasts
- Conducted financial data analysis and optimized machine learning pipelines to enhance investment decision accuracy
- Collaborated with financial analysts to refine forecasting models and incorporate domain insights
- Documented model architecture, assumptions, and data preprocessing workflows in project documentation
- Evaluated model performance using R^2 score and mean squared error to achieve an 87% accuracy on validation data

PROJECTS

Note Genie – Note Taker

September 2024

- Developed a full-stack web application using Node.js and Django to perform real-time speech transcription via Deepgram's API
- Implemented automated text summarization that converts transcribed audio into concise, easily digestible summaries
- Integrated third-party APIs to generate multiple-choice quiz questions based on the summarized content and visualize it through interactive mind maps using GoJS

ASL Gesture Recognition and Translation

September 2023

- Built a real-time hand gesture recognition app using OpenCV, TensorFlow/Keras, and Tkinter, integrating a CNN-based classifier and live webcam feed for dynamic gesture prediction.
- Automated gesture-to-speech conversion with gTTS and optimized computer vision pipelines for frame capture, preprocessing, classification, and GUI updates.

Real Time Chat Application

May 2023

- Developed a 'Real Time Chat App' using Node.js integrated with Socket.io for instant messaging
- Used the React.js library for front-end website and the MongoDB framework to store user data